

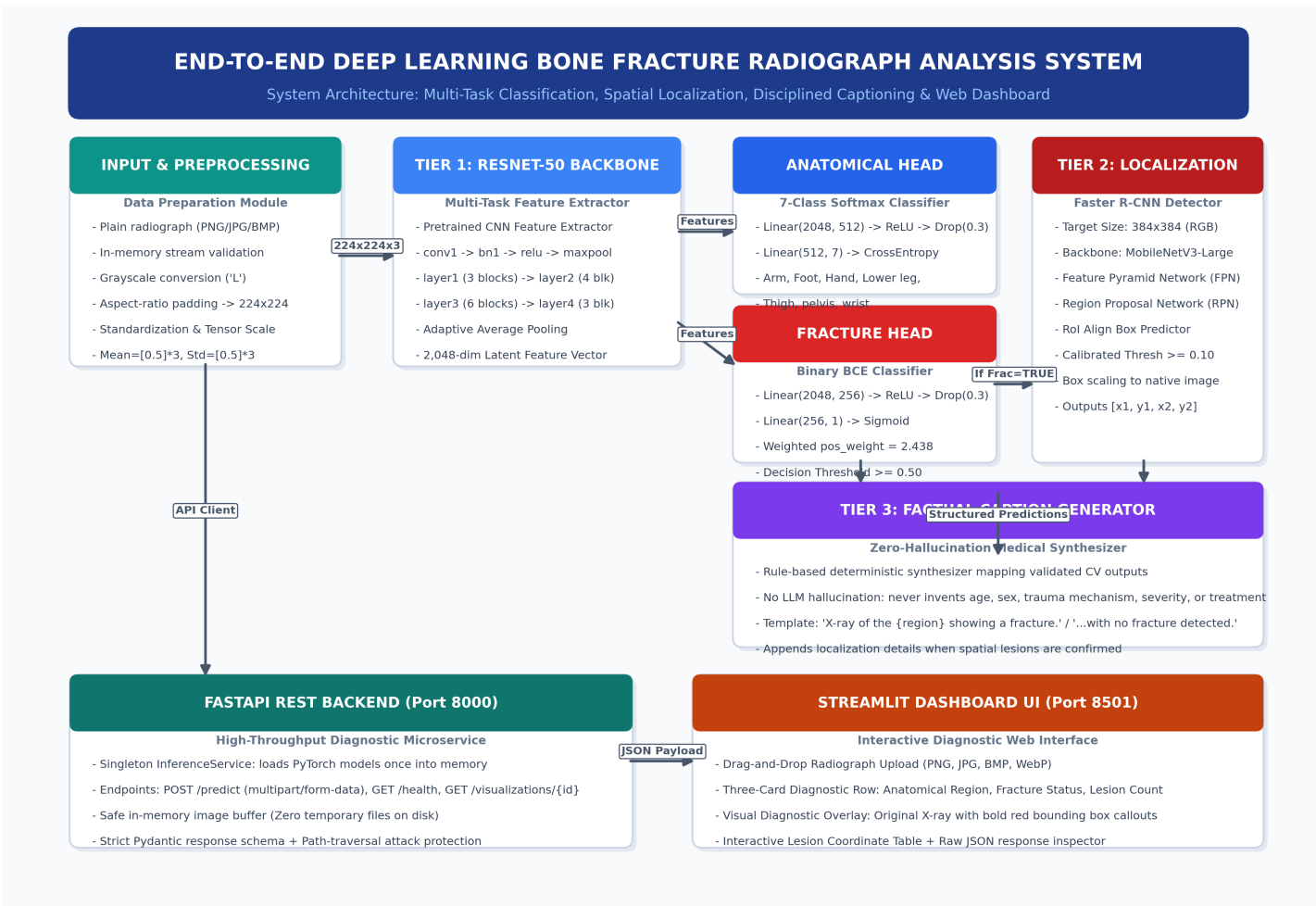
Bone Fracture Detection & Captioning System

System Architecture Specification & Technical Block Diagram Document | Version 1.0

1. System Architectural Overview

The Bone Fracture Image Captioning and Detection platform provides automated, multi-task radiograph analysis consisting of three primary execution tiers: (1) Multi-Task Shared Convolutional Backbone for 7-class anatomical site classification and binary fracture detection, (2) Faster R-CNN with Feature Pyramid Network (FPN) for spatial lesion localization, and (3) Disciplined Factual Caption Synthesizer generating non-hallucinatory medical text.

2. System Architecture Block Diagram



3. Deep Learning Component Specifications

Component	Architecture	Loss / Objective	Output Specification
Classifier Backbone	ResNet-50 (Shared)	Two-stage Transfer Learning	2,048-dim feature representation
Anatomical Head	2-Layer MLP + Drop(0.3)	Cross-Entropy Loss	7 classes: Arm, Foot, Hand, Lower leg, Thigh, pelvis, wrist
Fracture Head	2-Layer MLP + Drop(0.3)	BCEWithLogits (pos_weight=2.438)	Binary fracture probability (Threshold >= 0.50)

Spatial Detector	Faster R-CNN (MobileNetV3 FPN)	RPN + RoI Box Regression	Bounding boxes [x1, y1, x2, y2] (Thresh >= 0.10)
Caption Generator	Template Synthesizer	Deterministic rule engine	Factual natural language description

4. End-to-End Dashboard Interaction Workflow

- Step 1: Upload** — Doctor or clinical user uploads a plain radiograph into the Streamlit dashboard.
- Step 2: Analysis Trigger** — The 'Analyze X-ray' button dispatches an asynchronous multipart request to FastAPI POST /predict.
- Step 3: Multi-Task Classification** — ResNet-50 predicts the anatomical site and fracture probability.
- Step 4: Spatial Localization** — If fracture is positive, Faster R-CNN detects lesion bounds and generates a visual diagnostic overlay.
- Step 5: Visual Presentation** — Dashboard renders three KPI cards, factual caption, bounding box overlay, and coordinate tables.